

YZV475E - Data Visualization

2025 - 2026 Spring (CRN: 24010)

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Office hours: Monday 15.30-16.30, Thursday 12.30-13.30 at Informatics Institute room 212

Recommended Textbooks:

- Information Visualisation: An Introduction, R. Spence, Third Edition, Springer.
- A Primer on Making Informative and Compelling Figures, C. Wilke, Amazon O'Reilly, 2019.
- Storytelling with Data: Let's Practice! Cole Nussbaumer Knaflc Wiley, 2019
- Tableau Dashboard Cookbook, Packt Publishing, Jen Stirrup, 2014
- Matplotlib 3.0 CookBook Packt Publishing by Srinivasa Rao Poladi

Additional Resources: <https://bioinformatics.ccr.cancer.gov/docs/data-visualization-with-r/>

Grading

- Assessment Components:
 - Midterm exam (20%)
 - Homeworks (2) %20
 - Term project (20%)
 - Final exam (40%)

Homeworks and midterm will be take-home assignments.

- VF Conditions:
 - %70 Attendance

Weekly Schedule

We ek	Date	Lecture	Practice	Assignments
1	Feb12	Fundamentals of data visualization and storytelling concept		
2	Feb19	Data transformation, data quality and filtering for visualization for visualisation	PANDAS , Tableau, PowerBI	
3	Feb26	Visualisation for one or two dimensional data: histograms, bar charts, density plots and distributions	Matplotlib ,seaborn,plotly	
4	March5	Visualisation for multi-dimensional data: scatter plots, correlation plots	Matplotlib ,seaborn,plotly	Homework1-will be released
5	March12	Visualizing many distributions at Once, Grouping and clustering multi-dimensional data	Seaborn,plotly	Homework1-Deadline
6	March26	Interactive data visualisation	Dash-Plotly ,Tableau,PowerBI	Midterm will be released
7	April2	Visualization with dashboards	Tableau , Seaborn	
8	April9	Visualisation on hierarchical data	Tableau	Midterm Deadline
9	April16	Visualising geospatial data	PowerBI	Homework2-will be released
10	April23	Public Holiday		
11	April30	Visualising time series data	PowerBI	
12	May7	Visualising networks	Networkx	Homework2-deadline
13	May14	Student project presentations and feedbacks		Project Presentations
14	May21	Student project presentations and feedbacks		Project Presentation

Note: Programs shown as bold in the practice column will be the major application in that week

Term Project

- The project could be done as a team of 2-3 students.
- The subject of the project will a public data set selected by the students themselves. The same data set will not be used by more than one group (there will be a shared online list of selected datasets – please check it before you finalize your dataset).
- One report should be given as a team.
- Presentations of the project will be done with short presentations of 10 minutes per group.

Project Schedule

- *Week 5 (Mon by 10:00)*: Students submit their project proposal forms to the Ninova system. For teams with multiple members, each student must submit a separate form. Please use the proposal template on Ninova.
- *Week 6*: Project proposal review results are announced. Revise your project by Friday 17:00 of the same week, if requested by your TAs. Get started working on your projects.
 - Groups who fail to submit revisions by the end of Week 5 (Friday 17:00) will get 10% deduction on their project grade.
- *Week 9*: Make an appointment with one of your TAs to review your project progress so far during this week. Please make sure that you have reasonable progress to report by this time.
- *Week 14 (Wednesday 09:30)*: The reports and presentations should be uploaded to Ninova.
- *Weeks 14*: Project presentations

Project objectives

- Data analysis: Analyze the data to identify trends, patterns, and insights. Use data cleaning/filtering/transformation techniques learned in the course.
- Create a dashboard containing various visuals. Each dashboard should contain at least **three** of the following visual types:
 - Time series plot
 - Geospatial figure
 - Graph visualization
 - Distribution plot
 - Hierarchical data
 - Scatter plot
 - heatmap

Project Grading

- Each team member gets an individual grade from presentation (50%) and a report grade (50%).
- Project assessment criteria will be announced on Ninova.

Important Notes

- You have to follow the course announcements on Ninova and your grades on Kepler.
- Course related e-mail notifications will be sent to your ITU email account. Please check your emails on a regular basis.
- When sending an e-mail to the instructor or TA, use your ITU email accounts and always **include your full name and student number** at the end of the message.
- Some announcements may be made in the class without any corresponding record on Ninova. Hence, you are expected to follow the in-class announcements as well. If you miss a class, make sure that you talk to your classmates to see if there were any announcements that you need to know.
- Any form of cheating or plagiarism will not be tolerated. This includes actions such as, but not limited to, submitting the work of others as one's own (even if in part and even with modifications), providing work for others to submit and copy/pasting from other resources (including Internet pages, even if attributed). Serious offenses will be reported to the faculty administration for disciplinary measures.